# Group Project Final Report

Counting Reports or Counting Evidence: What Provenance
an LLM Orchestrator Needs to Discount Correlated Agent Reports

Group ID: 14

DSA4213: Natural Language Processing for Data Science
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## Administrative Information


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| **Project direction** | Provenance-aware aggregation of agent reports |
| **Repository** | https://github.com/JasonOng0109/dsa4213-agent-project |


## Project Summary


When workers in a multi-agent system share a retriever or a tool, the orchestrator that combines their reports can receive one document as several agreeing reports. We ask what the message between agents must disclose before the orchestrator's belief and decision follow the evidence. In 20,280 trials on 40 cases with invented facts, three judge models with reasoning off read fixed, reworded reports under conditions that differ only in what the judge is told. With nothing disclosed, two reports of a weak source beat one report of a strong source in 454 to 479 of 480 trials per judge. Told that two reports share a source, every judge raised its stated probability of the truth in at least 37 of 40 cases, yet stayed wrong in most trials. Told which source is the primary record, every judge was right in every trial. Adding "somewhat more reliable" to the first note gave 331 right answers of 360 on ten cases. Every note was true, so these results do not cover false notes.


## 1 Problem, Research Question and Motivation

Many language-model systems split a task across worker agents and give one model, the orchestrator, the job of combining their reports. Workers in the same framework often share a retriever or a tool, so their reports can be correlated. Consider three workers that receive the same document and each summarise it in their own words. The reports look like three witnesses, but only one source exists. An orchestrator that counts them as three independent confirmations grows confident on the strength of one source.

A multi-agent system holds the information that could prevent this failure. One model that reads three articles copied from one press release has no record of where they came from, while a multi-agent system knows which agent called which tool and which document came back. That record is lost when the workers' outputs are rendered as text for the orchestrator. Builders of these systems, this study's intended users, can keep part of that record as a field of the message between agents. Each field takes work to add, and to the best of our knowledge, no benchmark says which field changes what the orchestrator does. Our research question is the following.

*Can the orchestrator distinguish the number of agreeing reports from the number and quality of the independent sources behind them, and what must the inter-agent message tell it (a source ID, a source type, that reports share a source, or which source is more reliable) before its belief and its decision follow the evidence?*

We isolate this aggregation step as one controlled call, in which a judge model plays the orchestrator and reads fixed reports. "Belief" means a judge's stated probability, which may be uncalibrated or differ from its internal state. We defined success before collecting data. Each primary hypothesis had to show its predicted effect across judges and cases or be ruled out, so a negative result counts.

The TA asked why one call on fixed text is a multi-agent question. A multi-agent system is where the source record exists and gets lost, and we study the last step of that path, so we reworded the research question to name that step and the lost record. The design did not change, and Appendix D lists the rule changes made after a five-case pilot.

## 2 Related Work and Contribution

Xu et al. [14] separate conflicts within supplied evidence from conflicts with a model's own knowledge. Our setting is the first kind. ClashEval studies the second [13], which our invented entities rule out. Schuster et al. [10] find a credibility hierarchy that repetition can reverse, but their repeated evidence is verbatim, which deduplication would remove. RAMDocs gives each document its own agent, so count and independence cannot be separated [11]. AuthorityBench studies authority perceived from URLs [16].

Multi-agent debate treats agreeing outputs as corroboration [3]. Minority Sentinel trains a classifier to overturn wrong majorities [6], and conformity and sycophancy can push debate toward consensus [1, 15]. Our agents never interact, and their agreement comes from the shared source. Provenance-aware memory methods correct the count outside the model [8, 9], and tracing provenance through paraphrases remains open [12]. We measure what the judge does with no disclosure and with a one-sentence note.

Beyond using existing models and tools, we make four contributions. We compare disclosures on identical frozen reports, so the reports themselves cannot explain a difference between conditions. We measure belief and decision together, with an abstention arm. We build a corpus of separately generated paraphrases that carry no provenance. From work on correlation neglect, the failure to account for repeated information [2, 4], we adapt an index of how many independent sources a judge acts as if it saw.

## 3 Agent and Task Setup

A reporter model, Gemma 4 31B, plays the worker agents. It rewrites each source's content sentence as a report and never sees who wrote the source. A judge model plays the orchestrator. Each trial's prompt holds a question, four numbered options, the reports and any note. The judge returns an answer and an integer probability for each option, and the probabilities sum to 100. A message is one labelled report line in that prompt, and no tool is called (see Appendix E).

The judges are Claude Opus 5, Claude Sonnet 5 and GPT-5.4, called through OpenRouter and pinned to one serving provider each. Reasoning was switched off explicitly, and every main-run

trial logged zero reasoning tokens. The runner sends temperature 0, but no judge endpoint accepts it, so each judge ran at its provider's default.

A run stops itself on its call ceiling, a failed call, a change of provider, repeated parse failures or a leak of provenance into a report. One run stopped after 762 calls on an overload error inside a successful response, so we added a retry and reran it in full.

# 4 Study Design

Figure 1 shows the pipeline. Reports are generated once and reused in every condition.

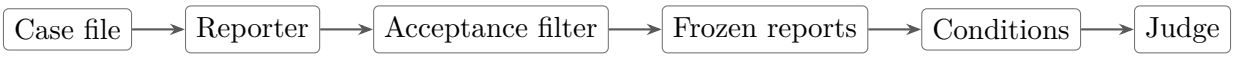



**Figure 1:** The study pipeline, from the authored case files to the judge's reply.

**Cases and frozen reports.** Each case fixes an invented truth, three strong sources B, C and D that state it by separate routes, a weak source A with a wrong value, and two distractors that no source supports. Four options separate a judge that follows A's claim from one that answers at random. The 40 cases cover institutional logistics (13), product specifications (13) and incident status (14), with ten per quality dimension (authority, directness, method and primary-source status). In 30 cases (T1), A's sentence carries a mild hedge that survives rewording. In the other ten (T2), A and B are equally flat assertions, so the tiers show whether a result needs the hedge.

A candidate report must keep its source's meaning and values (rated by Claude Sonnet 5), pass length and overlap checks, and contain no attribution. We froze 280 accepted reports and reused them across conditions. Team members reviewed the cases, and one member read every frozen report before judging. In seven cases, one of A's reports came out as the question turned into a statement, so we also report results without these echo cases (see Appendix A).

**Two complementary blocks.** In the agreement block, every report supports the truth. A0 has no report and A1 has one. A2 joins three rewordings of B in one message, A3 splits them into three lines from one agent [5], and A4 gives the lines three agent names. A5 adds a shared source ID, and A6 adds a note stating the dependence. A7 uses three independent sources and does not say that they are independent.

In the conflict block, two rewordings of weak A face one report of strong B, so counting reports gives the wrong answer. Table 1 lists what each condition discloses. We call X3's sentence the dependence note and X4's the quality note.

**Repeats and stages.** Stage 1 ran 17,760 trials on all 40 cases. Each judge had six repeats per agreement condition and 12 per conflict condition, and the 12 balance the truth's option position and the independent report's slot. An abstention arm adds the option "Cannot be determined from the evidence provided". Stage 2, two days later, ran 2,520 trials on ten preselected cases (three T2) that cover every domain and quality dimension. It ran the wording ladder W1–W3 for all three judges, and for Sonnet and GPT-5.4 it ran reports padded with neutral filler to 88–96 words, as a control for length [7]. The 20,280 trials exclude a 3,420-trial pilot on five of the 40 cases and the optional checks.

**Table 1:** Conflict conditions. All show the same three reports, and every disclosure is true.

| Condition | Information added to the reports |
|---|---|
| X1 | None. |
| X2 | Each report's source type, such as a student guide or an official register. |
| X3 | The two agreeing reports share a source, and B's source is independent. |
| X4 | B's source is primary and authoritative, and the others are secondhand or unofficial. |
| X5 | X3 plus a quality note that names the shared source as secondhand or unofficial. |
| W1–W3 | X3 plus "somewhat", "considerably" or "far more reliable" for B's source. |

**Measures.** We ranked the measures in advance, because they can disagree. Decision accuracy comes first, followed by decision stability, stated $P(\text{truth})$ and abstention. A confidence screen requires $P(\text{truth})$ to rise from an irrelevant report (P0) to a sentence ruling out two options (P1) to a report naming the answer (P2). Page's test for ordered alternatives checks this, with cases as blocks.

We also estimate how many independent sources a judge's stated probabilities imply. With $l(c)$ the logit of a case's mean stated $P(\text{truth})$ in condition $c$,

$$n_{\text{eff}}(c) = \frac{l(c) - l(\text{A0})}{l(\text{A1}) - l(\text{A0})}. \tag{1}$$

If each source carries equal weight and evidence adds in log-odds, a judge that knew the sources would give 1 for three rewordings of one source and 3 for three independent sources. We report medians over cases and treat the index as descriptive. It is undefined when one report moves the judge by less than 0.2 logits, and only a bound when a mean falls outside 1–99%.

**Hypotheses and analysis.** H1 predicts that repetition raises confidence and can beat one better-sourced report, and H2 that the rise comes from redundancy and segmentation more than agent names. H3 predicts that a dependence note shifts belief without changing decisions, and H4 that relative quality does change them. H5 predicts that named source types help, and H6 that a judge abstains where its probabilities do not back its answer. H3, H4 and H6 are primary, and Appendix B gives the criteria set before data collection.

A case's repeats reuse its reports, so they are not independent evidence. For each judge we therefore run one-sided sign tests over cases and bootstrap 95% intervals from 10,000 case resamples. These describe this corpus only, and a non-significant test does not establish equivalence.

## 5 Evaluation and Results

Where three numbers appear together, they are in the order Opus, Sonnet, GPT-5.4. All three judges passed the confidence screen (Page's $p < 0.0001$), so their stated probabilities can be ranked but are not shown to be calibrated. All 19 malformed stage-1 replies fell in A0, P0 or P1, and every stage-2 reply parsed. On average, the most common answer for a case and condition took 96–97% of its 12 repeats, against 41% by chance. A planned discrimination rule, dropped after the pilot because it measures how sure a judge is of A's claim, would have excluded Sonnet and GPT-5.4 from probability-based results (see Appendix D).

### 5.1 Repetition raised belief without adding sources

As Table 2 shows, three reports of one source raised every judge's belief. From A1 to A4, mean $P$(truth) rose in 40 of 40 cases for every judge and in all three domains (H1). For Opus and Sonnet, most of the rise came from splitting the rewordings into separate report lines, and naming three agents added less (H2). GPT-5.4 was near its ceiling from A2 on.

**Table 2:** Agreement block. Mean stated $P$(truth) (%) over 40 cases and six repeats per case. A1–A6 reuse one source, and A7 holds three independent sources.

| Condition | Opus | Sonnet | GPT-5.4 |
|---|---|---|---|
| A0: no report | 25.8 | 26.4 | 25.1 |
| A1: one report | 77.4 | 76.4 | 91.1 |
| A2: three rewordings, one message | 81.2 | 84.2 | 96.8 |
| A3: three report lines, one agent | 91.4 | 94.8 | 99.6 |
| A4: three report lines, three agents | 93.8 | 95.2 | 99.5 |
| A5: A4 plus shared source ID | 88.3 | 95.0 | 99.8 |
| A6: A5 plus dependence note | 85.6 | 91.3 | 99.6 |
| A7: three independent sources | 94.4 | 95.8 | 99.5 |

With the shared source hidden by the rewording, three reports of one source (A4) scored within a point of three independent sources (A7). Median $n_{\text{eff}}$ in A4 and A7 was 1.64 and 1.69 for Opus and 1.88 and 1.98 for Sonnet. GPT-5.4's index was at ceiling in 35 of 40 cases, and Opus never stated above 97 in A3, A4 or A7, which caps its index near 2, so neither can reach the 3 that independent sources would earn. With the dependence note (A6), away from the ceiling, three reports of one source still scored 1.24 for Opus and 1.52 for Sonnet.

### 5.2 The dependence note moved belief and few decisions

With nothing disclosed, the judges were right in 1, 2 and 26 of 480 trials, and every wrong answer named A's claim (see Table 3). The note in X3 raised mean $P$(truth) by 15.5 points for Opus (95% CI 14.8–16.3), 12.3 for Sonnet (11.7–13.0) and 7.3 for GPT-5.4 (5.7–8.7), in 40, 40 and 37 of 40 cases. Mean $P$(A's claim) fell only to 52.8, 58.3 and 60.3, still above $P$(truth) for every judge.

**Table 3:** Conflict block, over 40 cases and 480 trials per judge and condition.

| | Right, of 480 | | | Mean $P$(truth) (%) | | |
|---|---|---|---|---|---|---|
| | Opus | Sonnet | GPT-5.4 | Opus | Sonnet | GPT-5.4 |
| X1: no disclosure | 1 | 2 | 26 | 25.7 | 20.2 | 22.6 |
| X2: source types | 403 | 193 | 283 | 60.9 | 41.1 | 52.0 |
| X3: dependence | 63 | 8 | 27 | 41.2 | 32.5 | 29.9 |
| X4: quality | 480 | 480 | 480 | 68.9 | 74.3 | 82.0 |
| X5: both | 480 | 480 | 480 | 68.8 | 67.3 | 75.8 |

H3's belief half held for every judge, but its unchanged-decision half failed for Opus. Opus gained right answers in 14 cases and lost them in none ($p < 0.0001$). All 63 of its right X3 answers

fell in T1 cases, where A's hedge already leaned toward the truth. Sonnet rose from 2 right answers to 8 and GPT-5.4 from 26 to 27. Their sign tests ($p = 0.0625$ and $p = 0.377$) neither establish a rise nor rule one out. In the ten T2 cases, every judge scored 0 of 120 with or without the note.

### 5.3 A true quality note changed decisions

With the quality note (X4) or both notes (X5), every judge was right in all 480 trials (H4). Adding the dependence note to the quality note changed no decision and lowered mean $P$(truth) by 6 to 7 points for Sonnet and GPT-5.4. Source-type names (X2) also helped (H5), with 403, 193 and 283 right decisions. The gain remained without cases naming a familiar kind of public or official body. It was largest on method cases and smallest on primary-source cases, where Sonnet and GPT-5.4 were right in only 12 and 24 of 120 trials (see Appendix C).

Figure 2 shows the wording ladder, which softens the quality note. At "somewhat more reliable", Opus and GPT-5.4 were right in all 120 trials and Sonnet in 91. The judges chose the better source, yet mean $P$(truth) was only 50–54%. From "considerably" on, every judge was right in every trial. Mild wording, added to the dependence note, was enough to change decisions. Every note favoured the truth, however, so this result cannot rule out simple compliance.

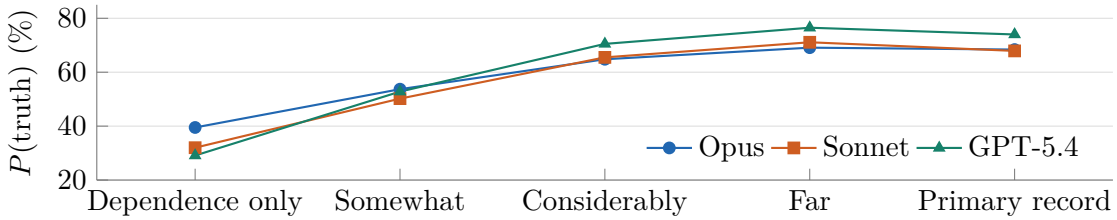



**Figure 2:** Mean stated $P$(truth) on the wording ladder, over ten cases and 120 trials per judge and condition. X3 and X5 come from stage 1. Error bars are omitted, as case-resampling intervals are at most ±2.0.

### 5.4 An abstention option rarely exposed uncertainty

H6 flags a forced-choice reply whose answer is not its top option or whose top-two margin is below ten points, after rescaling all four probabilities to sum to 100. Only 30, 3 and 7 of each judge's 2,400 conflict trials were flagged, all of them close calls. The abstention arm reruns four of the 12 repeats, so only six flagged trials had a matching abstention trial, and none of the six abstained.

All three judges can abstain. Shown one irrelevant report, they abstained in 160, 153 and 160 of 160 trials, and none abstained in A7. Yet Opus and Sonnet never abstained in their 800 conflict trials, and GPT-5.4 abstained in 9 of 800. Even in X3's T2 cases, where the note leaves one flat claim against another, none of the 120 trials abstained. H6 therefore found no support, and with only six direct tests, these rates carry the reading. With nothing disclosed, mean $P$(A's claim) was 69 to 71, so the judges had little doubt to express.

### 5.5 Comparisons and ablations

**Long reports.** With 88–96-word reports (Sonnet and GPT-5.4, ten cases), the conflict block broadly matched the short-report results, though source-type names helped less. GPT-5.4 was

right in 18 of 120 X3 trials against 3 with short reports, all 18 in T1 cases, as with Opus. Sonnet's A1-to-A2 rise was 13.4 points with long reports against 8.0 with short ones, larger in nine of ten cases ($p = 0.011$). The two lengths ran on different days, so this does not isolate a length effect.

**Optional checks.** Dropping the seven echo cases reversed no result. On a second report set for two cases, decisions in X1, X3, X4 and X5 did not change, and X2 rose for Sonnet and GPT-5.4. On two cases, Sonnet's decisions matched with reasoning on and off. Llama 4 Maverick and Qwen 3.6 27B, which return token probabilities unlike the main judges, ordered the screen's levels alike in stated and token form (Spearman 0.83 and 0.81, see Appendix C).

## 6 Analysis of Results, Failures, Limitations, Costs and Risks

Against the patterns fixed in advance, H1, H4 and H5 held, H2 held with short reports, H3's belief half held for every judge, and H6 found no support.

**What the message should carry.** Of the three fields tested in the conflict block, only a quality note made every judge right. A fourth field, an opaque source ID, ran only in the agreement block, where it lowered Opus's belief by 5.5 points and moved the others by 0.3 or less, so its effect on decisions is unknown. A dependence note costs little, because the system already records which reports share a source. A quality note needs a judgment of source quality, and that can be wrong. So the cheapest field changed few decisions, and the field that worked is the hardest to produce. Where relative source quality cannot be established reliably, a source type is a promising alternative. A system can record it, and it improved accuracy in this corpus, although its effectiveness varied substantially across judges and quality dimensions.

**Why the dependence note fell short.** Decisions followed each judge's top stated probability, since no forced-choice conflict answer sat below another option, though the prompt asks for the answer first, which may add to this link. The dissociation between belief and decision is therefore a difference of size. The dependence note raised $P$(truth) by 7 to 16 points, too little to pass $P$(A's claim), while the quality note raised it by 43 to 59. Taken at face value, X3's note turns two against one into one source against one, so a judge that discounted the duplicate fully would hold the two claims about equally. None did.

In an optional comparison on two cases, Sonnet wrote reasoning in 20 of its 48 X3 trials. Two Claude agents coded all 20 traces as noticing the shared source, a code that includes restating the note, and all 20 trials chose A's claim. In the trace below, the letters are agent slots. Agent A held the truth (65 Wh), and Agents B and C held the weak source's claim.

> *B and C actually trace back to the same source, so it's really just A's 65Wh versus one independent claim of 55Wh rather than two separate confirmations. Still, since two agents mention 55Wh even if derived from one source, I'll nudge my probability slightly toward 55Wh being correct.*

Sonnet then stated 55% for 55 Wh and 35% for the truth. With both notes (X5) it was right in all 48 trials, and a trace on the same case, where Agent B held the strong report, ends "That means I should weight B more heavily". These traces come from one judge, so they say nothing about the others. In A6 and X3, repetition kept some weight after the dependence was declared, as in correlation neglect in people [4].

**Limitations.** The judge reads fixed text in a single call, so we cannot say whether these notes improve real pipelines. Each agent reads exactly one source, every case is an invented single fact with four options, every conflict is two against one with the majority wrong, and every report comes from one reporter. The equivalence rater, Claude Sonnet 5, is also a judge, and two judges share a developer. Reasoning was off except in a two-case check, where Sonnet wrote reasoning in only 32 of 144 trials. No arm shows one A report against one B report, so the lean in X3 is not tied to repetition alone. Comparisons between stages carry day-to-day drift. The ladder and length arm used ten cases and the optional checks two to fifteen. Every main result comes from one prompt, and no control added neutral text of equal length.

**Costs.** Logged API spend came to US$52.19 over 27,768 calls, and Appendix D.1 itemises it. The 20,280 main-study trials took US$40.54 (about US$2 per 1,000) and 8.7 million tokens. Stage 1 ran in three parallel lanes of 18 calls a minute each, and the longest lane, the conflict block's 7,200 calls, took about 6.7 hours.

**Risks.** The judges followed a quality note even when it was mild and sat beside the dependence note. If a false note is followed as readily, whoever writes the field could steer the orchestrator's choice. A system that carries this field must verify it and take it from a trusted component, because a faulty estimate or a worker's self-rating could steer the decision. The numbers describe these model versions only, and hosted models change.

## 7 Reproducibility and Responsible Practice

The repository holds the code, case files, frozen reports, trial logs, setup instructions and exact model identifiers and provider settings. The study needs Python with `uv`. Checking the numbers needs only the committed logs, and rerunning the study needs an OpenRouter key.

The study has no human subjects and no personal data. Every entity is invented, so no output can misinform about a real institution, product or incident. The API key stayed in a local file that version control ignores. Claude Code was used extensively to record decisions, draft documentation and maintain the project's knowledge graph, and as an implementation assistant. The report was drafted with Claude's help, then proofread and edited by all team members.

## 8 Conclusion

We asked what an orchestrator must be told before it stops counting reports and starts counting evidence. For the three judges we tested, learning that two reports share a source was not enough, and learning which source is better was. Every note was true, so we cannot tell whether the judges weighed it or simply obeyed it. Where relative quality is independently known, the message should carry it, and that field must be correct, because the judges act on it. Future work should test false quality notes, mild notes alone, a source ID in the conflict block, neutral-text controls, real tool-call messages and several aggregation steps, with more reporters and judges.

## Contribution Statement

| Member (Student ID) | Main contributions | Share (%) |
|---|---|---|
| Ong Jia Xi (A0276092Y) | Led the project, built the study pipeline and ran the paid experiments. Completed final case reviews and merges, reviewed all frozen reports, and authored M1 and M38–M40. Wrote the stage reports and the first draft of the final report, revised it into the course template, and made the presentation's results slides. | 29% |
| Lee Kai Rong (A0287723U) | Authored nine cases (M5 and M30–M37) and reviewed 17 cases (M4, M8–M9, M16–M19, M22–M28 and M38–M40). Coordinated the corpus and ten-case subset, led the discrimination pass-mark decision, and raised post-pilot changes to the screen, ceiling rule and H6 interpretation. Supported drafting the final report. | 22% |
| Kathleen Pang Qi Yu (A0282219A) | Authored M4 and M22–M29 and reviewed M3, M14–M21 and M38–M40. Wrote the hedge-counting tool and drafted the revised research question. | 17% |
| Ee Rui Kai Rachel (A0281201X) | Authored M3 and M14–M21 and reviewed M2 and M6–M13. Led the wording-ladder sentence review and suggested refinements to the final report's methodological interpretations and phrasing. | 17% |
| Crystal Lai Qiaohui (A0282431L) | Authored M2 and M6–M13 and reviewed M5, M30 and M31. Led the H6 margin-cutoff decision. | 15% |
| Total | | 100% |

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# A Detailed protocol and experimental design

## A.1 Corpus construction and report acceptance

Five members wrote the cases under 20 authoring rules. These require comparable specificity and scope, neutral question wording, no time markers, provenance kept apart from the content sentence, and separate evidential routes for B, C and D. From M6 on, A's and B's type names differ only in the case's quality dimension, and the five pilot cases keep their original names. A static linter checked the rules a script can test, and every case from M2 on had a reviewer other than its author.

Each round of report generation sampled six candidates at temperature 0.9, rising to 0.95 in a second round. A candidate had to satisfy all four acceptance requirements:

1. Claude Sonnet 5 marked all details retained, nothing added, all values unchanged, and scope or denominator unchanged, with an equivalence score of at least 80/100.
2. Word-bigram Jaccard overlap was at most 0.70 with the source, and at most 0.50 with another retained report of that source or a source appearing in the same trial.
3. The report contained neither a word unique to its source's type name nor an attribution phrase such as "according to".
4. Its length differed from the source sentence by no more than three words.

Failed candidates were regenerated and never edited by hand, and generation stopped if a source lacked reports after two rounds. The provenance-leak check ran again over the frozen reports before judging.

Generation used 1,956 calls for 160 sources. Of 972 candidates, 721 passed the checks made on each candidate alone. We kept 280 reports, two for A, three for B and one each for C and D per case. The other 441 were surplus or too close to a report already chosen, and only two sources needed a second round. Rejection counts, which can overlap, were 105 for length, 72 for overlap with the source, 66 for overlap with another source's report and 12 for equivalence. In the reading before judging, all T1 hedges survived and no case was excluded. The seven echo cases are M3, M9, M13, M23, M28, M38 and M40. We leave them out only as a sensitivity check.

## A.2 Condition definitions and counterbalancing

The agreement block repeats strong source B, so any change shows up in $P(\text{truth})$. Source letters A–D name authored sources, while the displayed agent letters name rotating report slots. Table 4 lists every condition, and Table 5 lists what ran.

**Table 4:** The 13 conditions and the wording ladder. Agent labels (Agent A, B, C) name report slots and rotate with the order, and they are unrelated to the source names A–D. A message is one "Agent ... says" line of the single prompt. Appendix E shows example prompts in full.

| | Reports | Messages | Agents | Sources | What the judge is told besides the reports |
|---|---|---|---|---|---|
| A0 | 0 | 0 | 0 | none | nothing. The question alone gives the judge's prior |
| A1 | 1 | 1 | 1 | B | nothing |
| A2 | 3 | 1 | 1 | B | nothing. Three rewordings of B are joined into one message |
| A3 | 3 | 3 | 1 | B | nothing. Lines labelled "Agent A, report 1" to "report 3" |
| A4 | 3 | 3 | 3 | B | nothing |
| A5 | 3 | 3 | 3 | B | an opaque source ID, "(source: S-001)", on every report |
| A6 | 3 | 3 | 3 | B | A5, plus a dependence note: all "obtained their information from the same underlying source (S-001)" |
| A7 | 3 | 3 | 3 | B, C, D | nothing |
| X1 | 3 | 3 | 3 | A, A, B | nothing |
| X2 | 3 | 3 | 3 | A, A, B | each report's source type name |
| X3 | 3 | 3 | 3 | A, A, B | a dependence note: the two A reports "obtained their information from the same underlying source", while B's "comes from a different, independent source" |
| X4 | 3 | 3 | 3 | A, A, B | a quality note: B's source "is a primary, authoritative record; the other reports rest on secondhand or unofficial material" |
| X5 | 3 | 3 | 3 | A, A, B | X3's note, plus a quality note: B's source "is a primary, authoritative record; the source shared by" the other two "is a secondhand or unofficial one" |
| W1–W3 | 3 | 3 | 3 | A, A, B | X3's note, plus a softened quality note: B's source "is somewhat / considerably / far more reliable than the source shared by" the other two |

The conflict block's 12 repeats pair every answer position with every independent-report slot once, and every answer position with every wrong-claim offset once. Agreement uses six repeats under the same fixed schedule in every condition. The abstention arm's four repeats put the truth in each answer position once. These repeats measure response stability and reduce position imbalance.

The length arm embeds each report in one of three authored sets of neutral filler. Filler is assigned by slot and mentions no answer option. Each long report has 88–96 words, the longest within a trial is at most 1.10 times the shortest, and the report sentence rotates among the start, middle and end. The arm uses Sonnet and GPT-5.4, one judge per developer. The ten-case subset is M1–M6, M14, M22, M30 and M38. Under a selection rule that predates the pilot, every domain appears at least three times, every quality dimension at least twice, and three cases are T2.

The abstention arm adds "Cannot be determined from the evidence provided" to X1–X5, A7 and P0. P0 is an irrelevant report that checks whether the judge can abstain. A7 checks whether it abstains despite three agreeing independent sources. Before the main run we fixed two labels. A judge that abstains in fewer than half of its P0 trials "abstains nowhere", and one that abstains in at least half of its A7 trials "abstains everywhere".

**Table 5:** What ran. The short arm and the length arm each have an agreement entry and a conflict entry. We fixed the design of each optional sub-experiment before it ran.

| Entry | Conditions | Repeats | Judges | Cases | Trials |
|---|---|---|---|---|---|
| *Main run, stage 1* | | | | | |
| Agreement | A0–A7 | 6 | all three | 40 | 5,760 |
| Conflict | X1–X5 | 12 | all three | 40 | 7,200 |
| Abstention | X1–X5, A7, P0 | 4 | all three | 40 | 3,360 |
| Confidence screen | P0, P1, P2 | 4 | all three | 40 | 1,440 |
| *Main run, stage 2* | | | | | |
| Wording ladder | W1–W3 | 12 | all three | 10 | 1,080 |
| Length, agreement | A1, A2, long | 6 | Sonnet, GPT-5.4 | 10 | 240 |
| Length, conflict | X1–X5, long | 12 | Sonnet, GPT-5.4 | 10 | 1,200 |
| Main run in total | | | | | 20,280 |
| *Optional sub-experiments* | | | | | |
| Reasoning off and on | X1, X3, X5 (short, long) | 12 | Sonnet | 2 | 288 |
| Second report set | X1–X5 | 12 | all three | 2 | 360 |
| Stated vs. token probabilities | P0, P1, P2 | 4 | Qwen, Llama | 16[a] | 384 |
| Independence index, long | A4, A6, A7, long | 6 | Sonnet | 9 | 162 |

[a] We read 15 cases. M10's 24 trials were set aside because its option values are also option numbers, and M12 was judged in its place.

### A.3 Judging, validation, and analysis details

The judges were pinned to Claude Platform on AWS for Claude and to OpenAI for GPT-5.4. Records keep the raw reply, parse, provider, token usage and cost. The equivalence rater, like the judges, ran at its provider's default temperature. A six-call probe of identical prompts returned the same answer six times for Sonnet and five times for Opus and GPT-5.4, so responses were consistent, though not provably deterministic.

Stage 1's 18 imperfect records were 15 probability sums outside 90–110, two answers naming no option and one missing option probability. With the one parse failure, all were in A0, P0 or P1. No completed main-run call changed provider or used reasoning.

Decision stability is the share of a cell's repeats (one judge, case and condition) that give its most common answer. Under random answering it would be 0.41 for 12 repeats, 0.47 for six and 0.53 for four. Conflict stability averaged 0.96–0.97, all 12 answers agreed in 168–178 of the 200 cells per judge, and the A1–A7 values were 1.00.

For the confidence screen, each case contributes its mean over four rotations to Page's test, with a planned one-sided threshold of 0.05. P0 supplies irrelevant text, P1 rules out two options and keeps the truth, and P2 uses B's first frozen report. Within-condition discrimination was below 0.5 in X1 and X3 (AUROC 0.01–0.37), because the judges were surer of their wrong answers, which follow the two agreeing reports, than of their right ones. X2 AUROCs were 0.71, 0.37 and 0.57 for Opus, Sonnet and GPT-5.4.

Equation (1) adapts correlation-neglect reasoning [2, 4]. Its reference values of one for repeated reports and three for independent sources assume additive log-likelihood contributions of equal strength, a common prior and a faithful probability scale, so they are benchmarks under that model. We clip means outside 1–99% and report them as bounds. Ceiling flags affect 0, 1 and

35 cases for Opus, Sonnet and GPT-5.4. Even without clipping, a compressed scale can hide differences. With Opus's mean A0 and A1 (25.8 and 77.4), three sources adding in log-odds would put $P$(truth) at 99.7, but Opus never stated more than 97 in any of its 720 A3, A4 and A7 trials, and its most common value was 94 in both A4 and A7. At 97 its index reaches only about 2, so the reference value of 3 for A7 was out of reach, and the small A7 − A4 gap is partly a ceiling effect. Sonnet stated 100 in about a quarter of its A4 and A7 trials (54 and 62 of 240).

Sign tests over cases count positive and negative paired changes and omit ties. The 95% bootstrap intervals resample whole cases 10,000 times with seed 4213. Reported means follow the scorer, which rounds within-case means to one decimal before averaging cases, so direct averages of raw trials can differ by 0.1 percentage points. Counts and conclusions are unaffected. We apply no correction for multiple comparisons.

A flagged H6 trial's abstention twin is the abstention-arm trial with the same case, condition and repeat. The planned contrast compares abstention in the twins of flagged trials with abstention in the twins of the others, within condition, combined with a Mantel–Haenszel difference and read alongside P0 and A7. Only six flagged trials have twins, four for Opus and two for GPT-5.4, so this contrast has little power to show the absence of an effect.

# B Hypotheses and interpretation

Table 6 gives the hypotheses in their wording from before data collection, and Table 7 sets each result beside the patterns fixed at that time. Those patterns set no number for an unchanged decision rate, so Section 5.2 reads H3 with sign tests over cases. H1 and H2 extend prior findings to paraphrased reports.

**Table 6:** The six hypotheses and the contrast that tests each. Section 4 defines the conditions.

| | Hypothesis | Tested by | Status |
|---|---|---|---|
| H1 | Correlated repetition raises the orchestrator's confidence and can beat a single better-sourced report | A1 → A4 and the X1 level | expected from prior work |
| H2 | The rise is driven by report redundancy and segmentation, not by distinct agent identities | A1 to A4 step by step and A4 against A7 | expected from prior work |
| H3 | Stating that two reports share a source shifts belief but does not change the decision | X1 → X3 | primary |
| H4 | Stating relative source quality does change the decision | X3 against X4 and X5, plus the wording ladder | primary |
| H5 | Judges infer quality from named source types without being told | X1 → X2 | secondary |
| H6 | Forced choice conceals belief. Where a judge's stated probabilities do not identify its answer, an explicit "cannot be determined" option will be taken when offered | the abstention arm | primary |

**Table 7:** Each hypothesis read against the patterns fixed before any data. A small or non-significant decision change does not establish equivalence.

| | Predicted evidence | Alternative pattern | Result |
|---|---|---|---|
| H1 | Repetition raises confidence. A repeated weak claim defeats a single strong claim. | An effect confined to one domain or report position. | **Support** in this corpus. A1 to A4 raises $P$(truth) in all 40 cases for every judge, and X1 is right in 1, 2 and 26 of 480. |
| H2 | A3 and A4 exceed A2 and differ little. | A larger A1 to A2 increase at long length. | **Support** with short reports, where segmentation outweighs agent identity. **Alternative pattern** for Sonnet with long reports (+13.4 against +8.0 points), though the comparison crosses days. GPT-5.4 has little headroom, and Opus was not tested. |
| H3 | Dependence disclosure raises $P$(truth) without changing decisions. | Decision changes occur only where a mild textual cue survives. | Belief shift supported for every judge. Decision effects remain inconclusive for Sonnet and short-report GPT-5.4. **Alternative pattern** for Opus (1 to 63 right, all in T1) and for GPT-5.4 with long reports (18 of 120 right, all in T1). |
| H4 | Quality disclosure changes decisions beyond dependence disclosure. | Decisions change only under the strongest wording. | **Support** for true notes. X4 and X5 are right in 480 of 480 for every judge, and "somewhat" in 120, 91 and 120 of 120. All 29 of Sonnet's wrong W1 answers named A's claim, 16 of them in the three T2 cases. The ladder rules out an effect of the strongest wording alone but cannot separate compliance from weighing the evidence. |
| H5 | Named source types improve accuracy. | Gains require a familiar real institution. | **Support**, varying by judge and dimension. Gains remain outside the eight familiarity-coded cases, though familiarity and authority are confounded. |
| H6 | Abstention concentrates where the judge's numbers do not support its answer. | A judge abstains indiscriminately or cannot abstain. | **No support**. None of the six flagged twins abstains, and GPT-5.4's contrast gives +0.00 on two twins. P0 and A7 show selective use of abstention, but six twins are too few to rule out the predicted relationship. |

# C Supplementary results and sensitivity checks

## C.1 Source-type effects and report length

Table 8 breaks X2 down by quality dimension. This breakdown is descriptive and sits outside the hypotheses. All judges did best on method cases and worst on primary-source cases. In eight cases, B's type names a public or official body that A's does not. In the other 32 cases, right decisions rose in 30, 19 and 25 cases and fell in none (one-sided sign tests, $p < 0.0001$). Seven of the eight familiar cases are authority cases, so familiarity and that dimension cannot be told apart. X2 also shows the same type name on both of A's reports, a hint that they share a source. Since X3's explicit dependence note moved few decisions, that hint is unlikely to explain the gain.

**Table 8:** Right decisions in X2 by the case's quality dimension, of 120 trials (ten cases, 12 repeats).

| | Authority | Directness | Method | Primary source |
|---|---|---|---|---|
| Claude Opus 5 | 106 | 93 | 117 | 87 |
| Claude Sonnet 5 | 56 | 59 | 66 | 12 |
| GPT-5.4 | 84 | 76 | 99 | 24 |

**Table 9:** Length-arm details on the ten-case subset. S and L denote short and long reports, judged on different days. Right decisions are of 120 per cell.

| | Sonnet S | Sonnet L | GPT-5.4 S | GPT-5.4 L |
|---|---|---|---|---|
| Right, X1 | 0 | 1 | 2 | 4 |
| Right, X2 | 52 | 36 | 87 | 74 |
| Right, X3 | 1 | 1 | 3 | 18 |
| Right, X4 | 120 | 120 | 120 | 120 |
| Right, X5 | 120 | 120 | 120 | 120 |
| Mean $P$(truth), X1 | 20.3 | 19.8 | 21.4 | 21.9 |
| Mean $P$(truth), X3 | 32.0 | 27.5 | 29.1 | 30.3 |
| Mean $P$(truth), A1 | 75.8 | 78.3 | 90.3 | 94.8 |
| Mean $P$(truth), A2 | 83.8 | 91.7 | 96.2 | 95.6 |
| A1 to A2 increase (points) | 8.0 | 13.4 | 5.9 | 0.8 |

Table 9 gives the length arm's details. GPT-5.4's long X3 beat its short X3 in seven cases and fell behind in none ($p = 0.008$). Sonnet's larger A1 to A2 rise holds in seven of eight cases without the subset's echo cases ($p = 0.035$). The rise fits a contribution from report length, but short and long prompts were judged on different days. On 24 identical A0 prompts shared by the pilot and main run, 22 decisions matched for each judge, and mean absolute $P$(truth) shifts were 0.5–2.7 points, with Sonnet at 2.7. These A0 comparisons do not directly estimate drift in the length contrast.

## C.2 Optional experiments and echo-case sensitivity

Table 10 lists these checks.

**Table 10:** Sensitivity and optional checks. Each finding is limited to the stated cases and models.

| Check | Evidence and qualification |
|---|---|
| Exclude seven echo cases | Every direction holds across 33 cases. X2 is right in 350, 156 and 243 of 396 for Opus, Sonnet and GPT-5.4. Without M3 and M38, Sonnet's A1 to A2 rise is 8.7 points short and 13.7 long. |
| Second report set, M1 and M2 | Every judge stays at 0 of 24 in X1 and X3 and 24 of 24 in X4 and X5. X2 stays at 24 for Opus and moves from 9 to 13 for Sonnet and from 22 to 24 for GPT-5.4. $P$(truth) moves by at most 7.5 points in any case and condition. The runs fell on different days. |
| Reasoning allowed, Sonnet, M1 and M2 | Decisions match reasoning-off trials at both lengths, with X1 and X3 right in 0 of 48 each and X5 in 48 of 48. Only 32 of 144 trials produce reasoning text, so the traces describe those trials alone. |
| Stated and token confidence, 15 cases | Llama 4 Maverick and Qwen 3.6 27B both order P0 $<$ P1 $<$ P2 in both measures (Page's $p < 0.0001$). Llama's trial-level Spearman correlation is 0.83 and Qwen's is 0.81. Within P1 they are 0.79 and 0.76. Qwen states 25% per option at P0, while its largest token probability averages 76.9%. |
| Long-report index, Sonnet, nine cases | Median $n_{\text{eff}}$ is 1.95 for A4 and 1.97 for A7 (short: 1.86 and 1.79 on the same cases). The median paired A4 minus A6 difference is 0.47 (short: 0.23). M38 is omitted because its long A7 construction falls one word below the band. Conditions span different days. |

The confidence cross-check ran 16 cases and uses 15. In M10 the answer values 2–5 are also option numbers, so a returned digit is ambiguous, and M12 took its place. The 15 cases give 360 trials, with token probabilities read at the answer digit.

Two fresh Claude agents coded the 32 nonempty reasoning traces independently, shuffled and blind to case, condition and result. They agreed on 31, and a team member settled the last, an X5 trace, as noticing the dependence. The X3 result does not depend on that call. The other 112 trials hold no reasoning text. Section 6 quotes trials `M2_X3_reasoning-on-short_run1` and `M2_X5_reasoning-on-short_run5`.

# D Deviations and costs

The pilot used M1–M5 and 3,420 judge trials. It was a design check and is not pooled with the main study. Table 11 lists what changed and what followed.

**Table 11:** Departures from the planned protocol and their effects.

| Original plan | What changed | Consequence |
|---|---|---|
| Ordering and AUROC above 0.5 gate stated probabilities | After the pilot and before the main run, only ordering decides. In conflict cells, discrimination measures how sure a judge is of the repeated wrong claim, which is the effect under study. | Sonnet and GPT-5.4 would have failed the original rule. Decisions are unaffected. |
| Temperature 0 for all judges | Temperature 0 is sent, but no judge endpoint accepts it. We found this after stage 1. | Provider defaults apply. Repeated calls and cross-day comparisons cannot be treated as deterministic or free of drift. |
| Three authored screen sentences | P0 uses filler, P1 an authored exclusion statement, and P2 B's first frozen report. | P2 still names the answer, but it is reporter output. |
| H6 interpreted against A7 alone | P0 and the selective-abstention cut-offs were added after the pilot and before the main run. | The controls establish that judges can abstain selectively, but six flagged twins still limit the direct H6 test. |
| Point estimates for the independence index | The scorer's original 1–99% clipping remains, but clipped means are reported as bounds following the pilot. | GPT-5.4's index is a bound in 35 of 40 cases. |
| Calibration summaries where spread permits | Brier scores were not computed. Analysis uses within-judge differences and ranks. | No hypothesis reading uses them. |
| Initial research-question wording | After the TA's feedback, the wording names the aggregation step and the lost source record. | The experimental design is unchanged. |

The stopped long-conflict log is kept and left out of scoring. In the second report set, two regenerated reports of D repeated existing wording, which cannot affect the check because it uses only A and B.

## D.1 API costs

Table 12 lists the logged spend. The total covers the pilot, the stopped first attempt and the optional experiments as well as the main study.

**Table 12:** Logged API spend in US dollars. Judge costs are the sum of each log's per-call costs, and generation costs are the fall in the account balance while generation ran.

| Item | Calls | Spent |
|---|---|---|
| Pilot, M1–M5: judging | 3,420 | 6.72 |
| P0 check after the pilot | 60 | 0.12 |
| Stage 1: generation and judging | 19,716 | 36.78 |
| Stage 2: judging | 2,520 | 5.25 |
| Stage 2: stopped first try of one entry | 762 | 1.60 |
| Optional sub-experiments, with 96 report calls | 1,290 | 1.72 |
| Total logged | 27,768 | 52.19 |

# E Prompt templates and logged replies

The A6 example gives the complete prompt. The following examples specify the changed content under the same template, with complete logged replies. The two block examples use case M1, repeat 1. The H6 mismatch example uses M20. Replies are copied from the trial logs. The system message for every judge call is "You are a careful, literal reader of the reports you are given."

## E.1 Agreement block: A6

This is trial `M1_A6_short_run1`. The truth is 15%.

```
You are asked to answer a factual question based on reports from others.
Read all reports below, then give your answer.

Question: What percentage of the final ME2103 grade comes from the lab logbook?
Possible answers:
  1. 15%
  2. 25%
  3. 10%
  4. 30%

Agent A (source: S-001) says: For module ME2103, the lab logbook contributes 15% toward the
↪  final grade.
Agent B (source: S-001) says: The final grade for module ME2103 includes a 15% contribution from
↪  the lab logbook.
Agent C (source: S-001) says: The lab logbook is worth 15% of the overall grade for module
↪  ME2103.
Note: Agent A, Agent B, and Agent C all obtained their information from the same underlying
↪  source (S-001).

Respond with ONLY a JSON object in exactly this format, nothing else:
{"answer": "<one of the possible answers, verbatim>",
 "p": {"<answer text>": <integer 0-100>, ...}}

"p" must give a probability for every possible answer listed above, using the
answer text as the key, and the values must sum to 100.

Do not include any explanation or reasoning -- output only the JSON object.
```

```
Claude Opus 5:   {"answer": "15%", "p": {"15%": 88, "25%": 4, "10%": 5, "30%": 3}}
Claude Sonnet 5: {"answer": "15%", "p": {"15%": 90, "25%": 4, "10%": 3, "30%": 3}}
GPT-5.4:         {"answer":"15%","p":{"15%":100,"25%":0,"10%":0,"30%":0}}
```

## E.2 Conflict block: X3, and what the other conditions change

This is trial `M1_X3_short_run1`. The truth is 15%, A's claim is 25%, and M1 is a T2 case. In this trial the Agent A slot holds the report of strong source B. The prompt is the same as above down to the options, and the reports and note are as follows.

```
Agent A says: The final grade for module ME2103 includes a 15% contribution from the lab
↪ logbook.
Agent B says: Twenty-five percent of the final module grade for ME2103 is derived from the lab
↪ logbook.
Agent C says: The lab logbook accounts for 25% of the final module grade for ME2103.
Note: Agent B and Agent C obtained their information from the same underlying source; Agent A's
↪ information comes from a different, independent source.
```

```
Claude Opus 5:   {"answer": "25%", "p": {"15%": 38, "25%": 57, "10%": 3, "30%": 2}}
Claude Sonnet 5: {"answer": "25%", "p": {"15%": 30, "25%": 60, "10%": 5, "30%": 5}}
GPT-5.4:         {"answer":"25%","p":{"15%":29,"25%":67,"10%":2,"30%":2}}
```

All three judges chose the repeated claim and put 29 to 38 on the truth. In the same repeat, the other conditions keep these reports and change only the following. X1 has no note. X2 has no note and labels each report with its source type.

```
Agent A (source: ME2103 assessment brief, section 4) says: ...
Agent B (source: student-run module wiki page) says: ...
Agent C (source: student-run module wiki page) says: ...
```

X4's note is "Note: Agent A's source is a primary, authoritative record; the other reports rest on secondhand or unofficial material." X5's note is X3's followed by "Agent A's source is a primary, authoritative record; the source shared by Agent B and Agent C is a secondhand or unofficial one." W1's note is X3's followed by "Agent A's source is somewhat more reliable than the source shared by Agent B and Agent C." W2 and W3 replace "somewhat" with "considerably" and "far". In the abstention arm, the following line precedes the final instruction.

```
If the reports do not determine the answer, you may instead answer exactly "Cannot be
determined from the evidence provided", still giving "p" over the numbered answers above.
```

### E.3 An answer whose own numbers favour another option

Trial `M20_P1_screen_run4` is the confidence screen's middle level, in the rotation that lists the truth last. The truth is 48 minutes. The prompt is the template above with the following content.

```
Question: What is the continuous runtime of the AeroGlide T5 vacuum cleaner on its standard
↪  power setting?
Possible answers:
  1. 41 minutes
  2. 34 minutes
  3. 55 minutes
  4. 48 minutes

Agent A says: The continuous runtime of the AeroGlide T5 vacuum cleaner on its standard power
↪  setting is between 37 minutes and 52 minutes.
```

```
Claude Opus 5:   {"answer": "41 minutes", "p": {"41 minutes": 40, "34 minutes": 5, "55 minutes": 5,
↪  "48 minutes": 50}}
Claude Sonnet 5: {"answer": "48 minutes", "p": {"41 minutes": 20, "34 minutes": 5, "55 minutes": 5,
↪  "48 minutes": 70}}
GPT-5.4:         {"answer":"48 minutes","p":{"41 minutes":21,"34 minutes":8,"55 minutes":4,"48
↪  minutes":67}}
```

The report leaves two options, 41 and 48 minutes, each four minutes inside a bound. Opus answered 41 minutes, the first of the two in the list, and then put 50 on the truth and 40 on 41 minutes. Its stated belief favours one answer and its decision names another, the strongest mismatch H6 describes. No such record occurred in the forced-choice conflict block.